

In[*]:= **(*PROBLEMA 6 INTERFERENCIAS*)**

In[*]:= **(*ÁLVARO MARTÍNEZ Y ANNA IRANZO, GRUPO BU2*)**

In[*]:= **n1 = 1;**

In[*]:= **n2 = 1.52;**

In[*]:= **$\lambda_0 = 550 * 10^{(-9)}$;**

In[*]:= **E0 = 1;**

In[*]:= **(* $\xi = k_0 z$ *)**

In[*]:= **$\xi_1 = -4 * \text{Pi}$;**
⌈número

In[*]:= **$\xi_2 = -\xi_1$;**

In[*]:= **(*APARTADOS A Y B*)**

In[*]:= **nA[ξ _] := n1 * HeavisideTheta[- ξ] + n2 * HeavisideTheta[ξ];**
⌈theta de Heaviside ⌈theta de Heaviside

In[*]:= **solA = NDSolve[{e'[ξ] + (nA[ξ])^2 * e[ξ] == 0,**
⌈resolvidor diferencial numérico
I * n1 * e[ξ_1] + e'[ξ_1] - 2 * I * n1 * E0 * Exp[I * nA[ξ_1] * ξ_1] == 0,
⌈número i ⌈número i ⌈ex...⌈número i
I * nA[ξ_2] * e[ξ_2] - e'[ξ_2] == 0}, e, { ξ , ξ_1 , ξ_2 }]
⌈número i

NDSolve: The scaled boundary value residual error of 1.781645388709543` indicates that the boundary values are not satisfied to specified tolerances. Returning the best solution found.

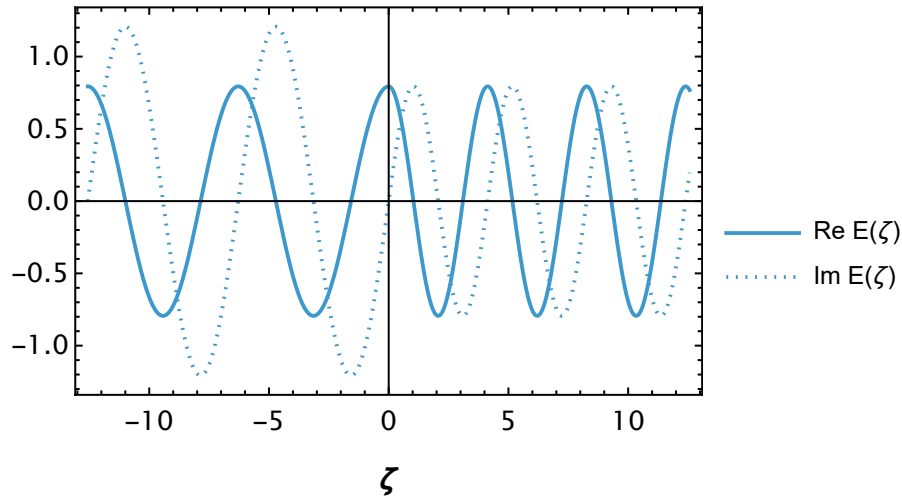
Out[*]=

{ { e → InterpolatingFunction[ Domain: {{-12.6, 12.6}} Output: scalar]] }

In[*]:=

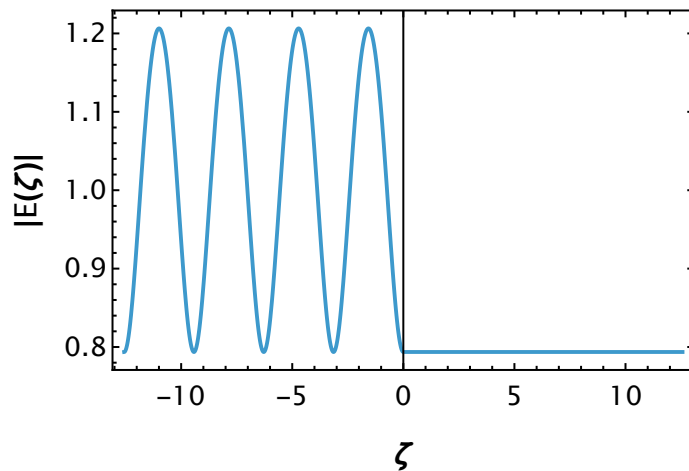
```
In[*]:= ReImPlot[Evaluate[e[ $\zeta$ ] /. solA], { $\zeta$ ,  $\zeta$ 1,  $\zeta$ 2},
  represent... [evalúa]
  PlotLegends → Automatic, ReImLabels → {"Re E( $\zeta$ )", "Im E( $\zeta$ )"},
  [leyendas de rep... [automático] [etiquetas de parte... [p... [número e] [p... [número e]
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  [estilo base] [familia de tipo de letra] [tamaño de tipo de letra]
  Frame → True, FrameLabel → Style[ $\zeta$ , 16, Bold]]
  [marco] [verd... [etiqueta de ma... [estilo] [negrita]
```

Out[*]:=



```
In[*]:= Plot[Abs[Evaluate[e[ $\zeta$ ] /. solA]], { $\zeta$ ,  $\zeta$ 1,  $\zeta$ 2},
  repr... [val... [evalúa]
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14}, Frame → True,
  [estilo base] [familia de tipo de letra] [tamaño de tipo de l... [marco] [verdad...
  FrameLabel → {Style[ $\zeta$ , 16, Bold], Style["|E( $\zeta$ )|", 16, Bold]}]
  [etiqueta de marco] [estilo] [negrita] [estilo] [número e] [negrita]
```

Out[*]:=



In[*]:= **(*APARTADO C*)**

```
In[*]:= rAnum = ((Evaluate[e[0] /. solA]) / E0) - 1
  [evalúa]
```

Out[*]:=

$\{-0.206349 - 1.5609 \times 10^{-7} i\}$

```

In[*]:= RAnum = Abs[rAnum]^2
          |valor absoluto
Out[*]=
{0.0425799}

In[*]:= RAteo = Abs[(n2 - n1) / (n2 + n1)]^2
          |valor absoluto
Out[*]=
0.04258

In[*]:= (*APARTADO D*)
          |deriv
In[*]:= n1 / n2
Out[*]=
0.657895

In[*]:= nL1 = 1.38;

In[*]:= nL2 = 1.69;

In[*]:= (nL1 / nL2)^2
Out[*]=
0.666783

In[*]:= (*APARTADO E*)

In[*]:= dL1 = 99.6 * 10^(-9);

In[*]:= dL2 = 81.4 * 10^(-9);

In[*]:= ξL1 = (2 * Pi / λ0) * dL1;
          |número pi

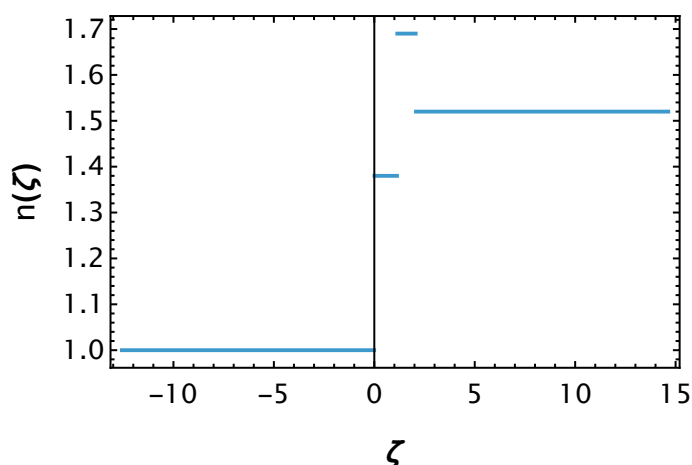
In[*]:= ξL2 = (2 * Pi / λ0) * dL2;
          |número pi
ξLE = (ξL1 + ξL2) + ξ2;

In[*]:= nE[ξ_] := n1 * HeavisideTheta[-ξ] + nL1 * HeavisideTheta[ξ] +
          |theta de Heaviside          |theta de Heaviside
          (nL2 - nL1) * HeavisideTheta[ξ - ξL1] + (n2 - nL2) * HeavisideTheta[ξ - (ξL1 + ξL2)];
          |theta de Heaviside          |theta de Heaviside

```

```
In[*]:= Plot[nE[ξ], {ξ, ξ1, ξLE}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  representación gráfica | estilo base | familia de tipo de letra | tamaño de tipo de letra
  Frame → True, FrameLabel → {Style[ξ, 16, Bold], Style["n(ξ)", 16, Bold]}]
  |marco |verd... |etiqueta de marco |estilo |negrita |estilo |negrita
```

Out[*]=



```
In[*]:= nE[ξLE]
```

Out[*]=

1.52

```
In[*]:= solE = NDSolve[{e''[ξ] + (nE[ξ])^2 * e[ξ] == 0,
  |resolvidor diferencial numérico
  I * n1 * e[ξ1] + e'[ξ1] - 2 * I * n1 * E0 * Exp[I * nE[ξ1] * ξ1] == 0,
  |número i |número i |ex... |número i
  I * nE[ξLE] * e[ξLE] - e'[ξLE] == 0}, e, {ξ, ξ1, ξLE}]
  |número i
```

⚠ NDSolve: The scaled boundary value residual error of 1.7017920823280865 indicates that the boundary values are not satisfied to specified tolerances. Returning the best solution found.

Out[*]=

```
{ {e → InterpolatingFunction[ Domain: {{-12.6, 14.6}} Output: scalar ] ] }
```

```
In[*]:= (*ReImPlot[Evaluate[e[ξ] /. solE], {ξ, ξ1, ξLE}, PlotLegends → Automatic,
  |represent... |evalúa |leyendas de re... |automático
  ReImLabels → {"Re E(ξ)", "Im E(ξ)"}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  |etiquetas de par... |p... |número e |p... |número e |estilo base |familia de tipo de letra |tamaño de tipo de le
  Frame → True, FrameLabel → Style[ξ, 16, Bold]] *)
  |marco |ver... |etiqueta de ... |estilo |negrita
```

```
In[*]:= plotn1 = ReImPlot[Evaluate[e[ξ] /. solE], {ξ, ξ1, 0}, PlotLegends → Automatic,
  |represent... |evalúa |leyendas de rep... |automático
  ReImLabels → {"Re E(ξ)", "Im E(ξ)"}, PlotRange → {{ξ1, ξLE}, Automatic},
  |etiquetas de parte... |p... |número e |p... |número e |rango de representación |automático
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  |estilo base |familia de tipo de letra |tamaño de tipo de letra
  Frame → True, FrameLabel → Style[ξ, 16, Bold]];
  |marco |verd... |etiqueta de ma... |estilo |negrita
```

```

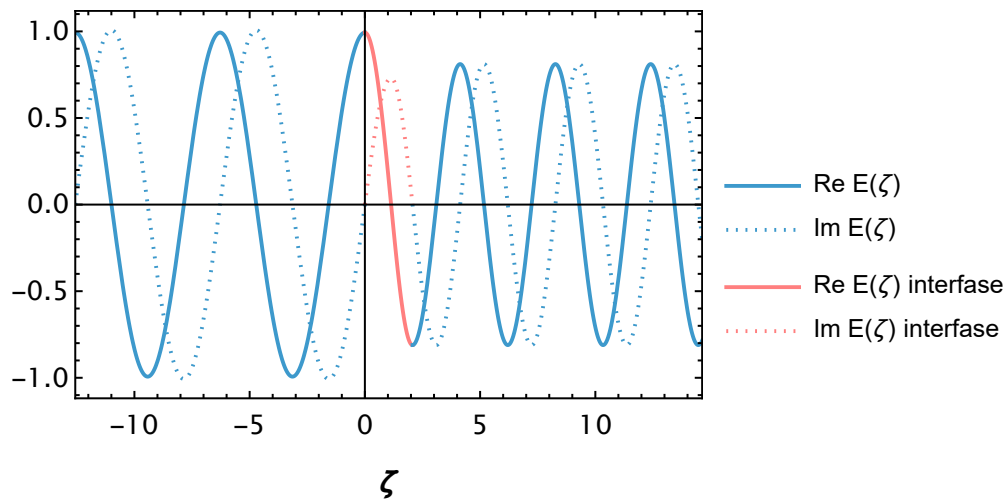
In[ ]:= plotn2 = ReImPlot[Evaluate[e[ξ] /. solE], {ξ, ξL1 + ξL2, ξLE},
  represent... evalúa
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  estilo base familia de tipo de letra tamaño de tipo de letra
  Frame → True, FrameLabel → Style[ξ, 16, Bold]];
  marco verd... etiqueta de ma... estilo negrita

In[ ]:= plotnL = ReImPlot[Evaluate[e[ξ] /. solE], {ξ, 0, ξL1 + ξL2},
  represent... evalúa
  PlotLegends → Automatic, ReImLabels → {"Re E(ξ) interfase", "Im E(ξ) interfase"},
  leyendas de rep... automático etiquetas de parte... p... número e p... número e
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14}, Frame → True,
  estilo base familia de tipo de letra tamaño de tipo de l... marco verdadero
  FrameLabel → Style[ξ, 16, Bold], PlotStyle → {Blend[{Red, White}], Thick}];
  etiqueta de ma... estilo negrita estilo de repre... mezcla... rojo blanco grueso

In[ ]:= Show[plotn1, plotnL, plotn2]
muestra

```

Out[]:=



```

In[ ]:= (*Plot[Abs[Evaluate[e[ξ] /. solE]], {ξ, ξ1, ξ2},
  repr... val... evalúa
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14}, Frame → True,
  estilo base familia de tipo de letra tamaño de tipo ... marco verdad
  FrameLabel → {Style[ξ, 16, Bold], Style["|E(ξ)|", 16, Bold]} *)
  etiqueta de ma... estilo negrita estilo número e negrita

In[ ]:= plotn1Abs = Plot[Abs[Evaluate[e[ξ] /. solE]], {ξ, ξ1, 0},
  repr... val... evalúa
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14}, Frame → True,
  estilo base familia de tipo de letra tamaño de tipo de l... marco verdadero
  FrameLabel → {Style[ξ, 16, Bold], Style["|E(ξ)|", 16, Bold]},
  etiqueta de marco estilo negrita estilo número e negrita
  PlotRange → {{ξ1, ξLE}, {0.7, 1.05}}, PlotLegends → {"|E(ξ)|"}];
  rango de representación leyendas de represe... número e

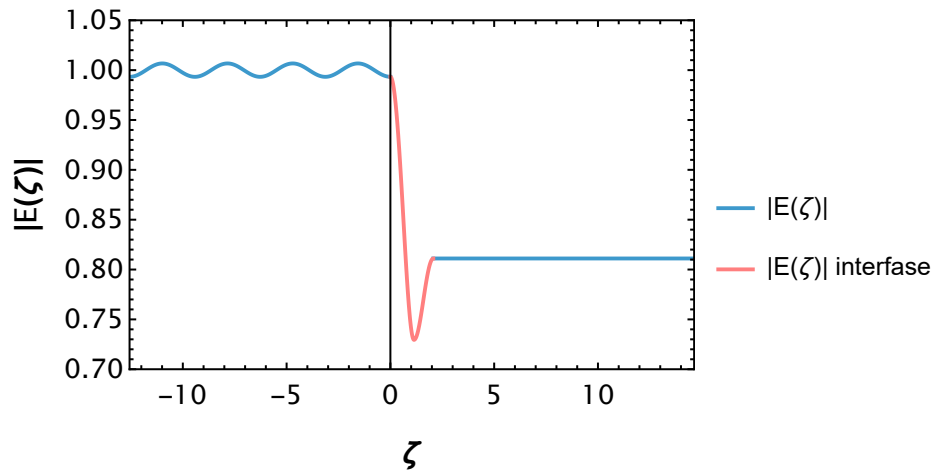
In[ ]:= plotn2Abs = Plot[Abs[Evaluate[e[ξ] /. solE]],
  repr... val... evalúa
  {ξ, ξL1 + ξL2, ξLE}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  estilo base familia de tipo de letra tamaño de tipo de letra
  Frame → True, FrameLabel → {Style[ξ, 16, Bold], Automatic}];
  marco verd... etiqueta de marco estilo negrita automático

```

```
In[*]:= plotnLAbs = Plot[Abs[Evaluate[e[ξ] /. solE]],
  {ξ, 0, ξL1 + ξL2}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  Frame → True, FrameLabel → {Style[ξ, 16, Bold], Automatic},
  PlotStyle → {Blend[{Red, White}], Thick}, PlotLegends → {"|E(ξ)| interfase"}];
```

```
In[*]:= Show[plotn1Abs, plotn2Abs, plotnLAbs]
```

Out[*]=



```
In[*]:= rEnum = ((Evaluate[e[0] /. solE]) / E0) - 1
```

Out[*]=

```
{-0.00670984 - 0.000106414 i}
```

```
In[*]:= REnum = Abs[rEnum] ^ 2
```

Out[*]=

```
{0.0000450333}
```

```
In[*]:= (*APARTADO G*)
```

```
ξLG[Ncel_] := Ncel * (ξL1 + ξL2) + ξ2;
```

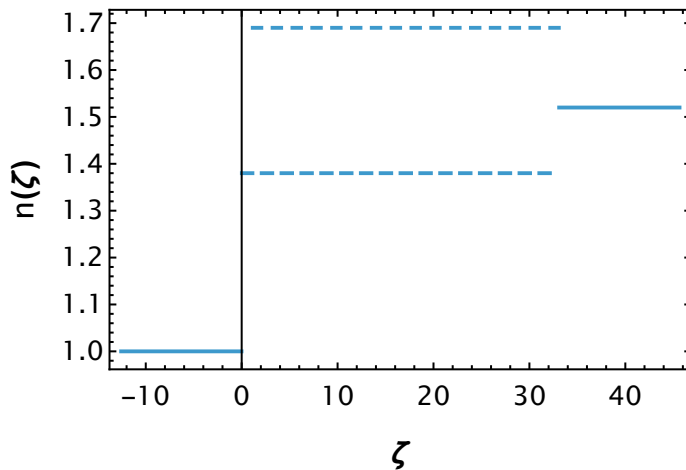
```

In[ ]:=  $\xi$ NL[Ncel_] := Ncel * ( $\xi$ L1 +  $\xi$ L2);
(*Para dos celdas*) nG[ $\xi$ _, Ncel_] := n1 * HeavisideTheta[- $\xi$ ] + nL1 * HeavisideTheta[ $\xi$ ] +
      (nL2 - nL1) * HeavisideTheta[ $\xi$  -  $\xi$ L1] + (nL1 - nL2) * HeavisideTheta[ $\xi$  - ( $\xi$ L1 +  $\xi$ L2)] +
      (nL2 - nL1) * HeavisideTheta[ $\xi$  - (Ncel *  $\xi$ L1 +  $\xi$ L2)] +
      (n2 - nL2) * HeavisideTheta[ $\xi$  - Ncel * ( $\xi$ L1 +  $\xi$ L2)];
(*Intento general*)
nGgen[ $\xi$ _, Ncel_] := n1 * HeavisideTheta[- $\xi$ ] + nL1 * HeavisideTheta[ $\xi$ ] +
      (nL1 - nL2) * Sum[HeavisideTheta[ $\xi$  - n * ( $\xi$ L1 +  $\xi$ L2)], {n, 1, Ncel}] +
      (nL2 - nL1) * Sum[HeavisideTheta[ $\xi$  - (n *  $\xi$ L1 + (n - 1) *  $\xi$ L2)], {n, 1, Ncel}] +
      (n2 - nL1) * HeavisideTheta[ $\xi$  - Ncel * ( $\xi$ L1 +  $\xi$ L2)];

Plot[nGgen[ $\xi$ , 16], { $\xi$ ,  $\xi$ 1,  $\xi$ LG[16]},
      BaseStyle -> {FontFamily -> "CambriaMath", FontSize -> 14}, Frame -> True,
      FrameLabel -> {Style[ $\xi$ , 16, Bold], Style["n( $\xi$ )", 16, Bold]}

```

Out[]=



```
In[*]:= (*Para Ncel=2*)
solG = NDSolve[{e'[ξ] + (nGgen[ξ, 2])^2 * e[ξ] == 0,
               I * n1 * e[ξ1] + e'[ξ1] - 2 * I * n1 * E0 * Exp[I * nGgen[ξ1, 2] * ξ1] == 0,
               I * nGgen[ξLG[2], 2] * e[ξLG[2]] - e'[ξLG[2]] == 0}, e, {ξ, ξ1, ξLG[2]}]
```

Out[*]=

```
{ {e → InterpolatingFunction[ Domain: {{-12.6, 16.7}} Output: scalar ] ] }
```

In[*]:=

(*APARTADO H*)

```
In[*]:= plotn1G = ReImPlot[Evaluate[e[ξ] /. solG], {ξ, ξ1, 0}, PlotLegends → Automatic,
                          ReImLabels → {"Re E(ξ)", "Im E(ξ)"}, PlotRange → {{ξ1, ξLG[2]}, Automatic},
                          BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
                          Frame → True, FrameLabel → Style[ξ, 16, Bold]];

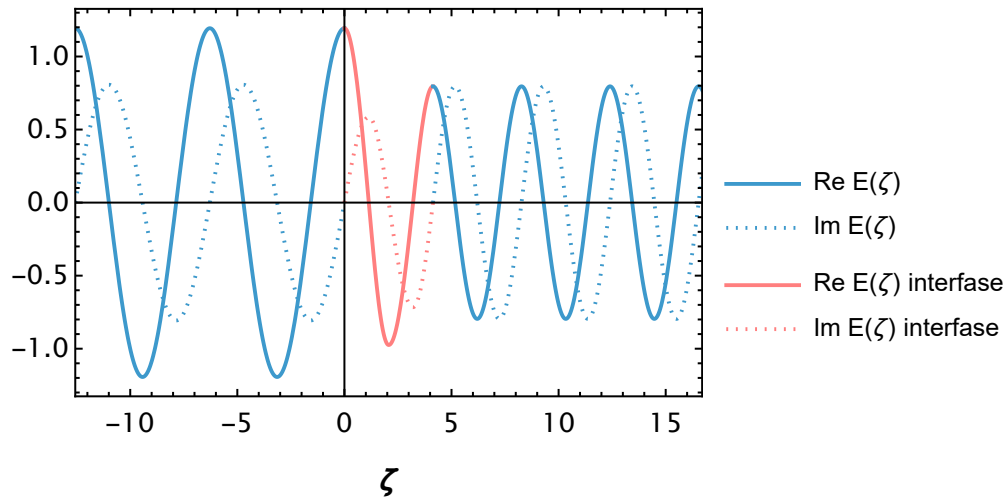
In[*]:= plotn2G = ReImPlot[Evaluate[e[ξ] /. solG], {ξ, ξNL[2], ξLG[2]},
                          BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
                          Frame → True, FrameLabel → Style[ξ, 16, Bold]];

In[*]:= plotnLG = ReImPlot[Evaluate[e[ξ] /. solG], {ξ, 0, ξNL[2]},
                          PlotLegends → Automatic, ReImLabels → {"Re E(ξ) interfase", "Im E(ξ) interfase"},
                          BaseStyle → {FontFamily → "CambriaMath", FontSize → 14}, Frame → True,
                          FrameLabel → Style[ξ, 16, Bold], PlotStyle → {Blend[{Red, White}], Thick}];
```



```
In[*]:= Show[plotn1G, plotnLG, plotn2G]
      muestra
```

```
Out[*]=
```



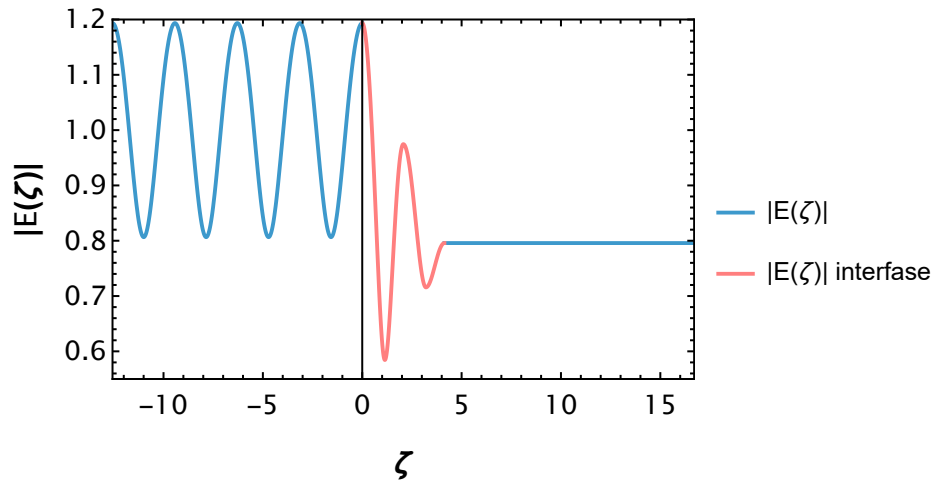
```
In[*]:= plotn1GAbs = Plot[Abs[Evaluate[e[ξ] /. solG]], {ξ, ξ1, 0},
      repr... val... evalúa
      BaseStyle → {FontFamily → "CambriaMath", FontSize → 14}, Frame → True,
      estilo base familia de tipo de letra tamaño de tipo de l... marco verdad...
      FrameLabel → {Style[ξ, 16, Bold], Style["|E(ξ)|", 16, Bold]},
      etiqueta de marco estilo negrita estilo número e negrita
      PlotRange → {{ξ1, ξLG[2]}, {0.55, 1.2}}, PlotLegends → {"|E(ξ)|"};
      rango de representación leyendas de represe... número e

In[*]:= plotn2GAbs = Plot[Abs[Evaluate[e[ξ] /. solG]],
      repr... val... evalúa
      {ξ, ξNL[2], ξLG[2]}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
      estilo base familia de tipo de letra tamaño de tipo de letra
      Frame → True, FrameLabel → {Style[ξ, 16, Bold], Automatic}];
      marco verd... etiqueta de marco estilo negrita automático

In[*]:= plotnLGAbs = Plot[Abs[Evaluate[e[ξ] /. solG]],
      repr... val... evalúa
      {ξ, 0, ξNL[2]}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
      estilo base familia de tipo de letra tamaño de tipo de letra
      Frame → True, FrameLabel → {Style[ξ, 16, Bold], Automatic},
      marco verd... etiqueta de marco estilo negrita automático
      PlotStyle → {Blend[{Red, White}], Thick}, PlotLegends → {"|E(ξ)| interfase"};
      estilo de repre... mezcla... rojo blanco grueso leyendas de represe... número e
```

```
In[*]:= Show[plotn1GAbs, plotn2GAbs, plotnLGabs]
[muestra]
```

```
Out[*]=
```



```
In[*]:= rHnum = (Evaluate[e[0] /. solG] / E0) - 1
[evalúa]
```

```
Out[*]=
```

```
{0.193466 - 0.000158824 i}
```

```
In[*]:= RHnum = Abs[rHnum]^2
[valor absoluto]
```

```
Out[*]=
```

```
{0.037429}
```

```
In[*]:= (*APARTADO I*)
(*N=4*)
```

```
solI4 = NDSolve[{e''[ζ] + (nGgen[ζ, 4])^2 * e[ζ] == 0,
[resolvedor diferencial numérico]
I * n1 * e[ζ1] + e'[ζ1] - 2 * I * n1 * E0 * Exp[I * nGgen[ζ1, 4] * ζ1] == 0,
[número i] [número i] [ex... [número i]
I * nGgen[ζLG[4], 4] * e[ζLG[4]] - e'[ζLG[4]] == 0}, e, {ζ, ζ1, ζLG[4]}]
[número i]
```

... **NDSolve:** The scaled boundary value residual error of 3.255565321893117` indicates that the boundary values are not satisfied to specified tolerances. Returning the best solution found.

```
Out[*]=
```

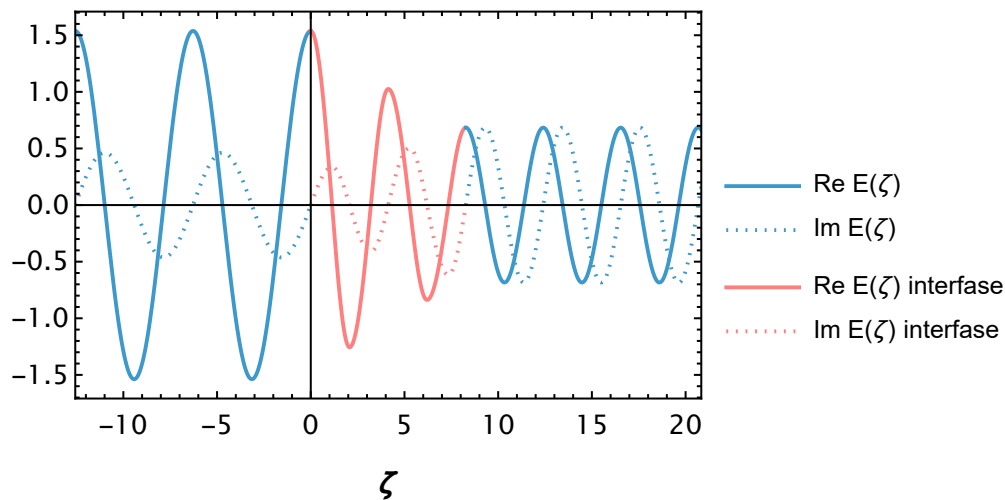
```
{ {e → InterpolatingFunction[
[+] [img alt="A small icon showing a graph of a function with a peak and a trough, representing an interpolating function." data-bbox="415 740 450 765"/> Domain: {{-12.6, 20.8}}
Output: scalar
] ] }
```

```

In[*]:= plotn1I4 = ReImPlot[Evaluate[e[ξ] /. soli4], {ξ, ξ1, 0}, PlotLegends → Automatic,
  represent... evalúa leyendas de rep... automático
  ReImLabels → {"Re E(ξ)", "Im E(ξ)"}, PlotRange → {{ξ1, LG[4]}, Automatic},
  etiquetas de parte... p... número e p... número e rango de representación automático
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  estilo base familia de tipo de letra tamaño de tipo de letra
  Frame → True, FrameLabel → Style[ξ, 16, Bold]];
  marco verd... etiqueta de ma... estilo negrita
plotn2I4 = ReImPlot[Evaluate[e[ξ] /. soli4], {ξ, NL[4], LG[4]},
  represent... evalúa
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  estilo base familia de tipo de letra tamaño de tipo de letra
  Frame → True, FrameLabel → Style[ξ, 16, Bold]];
  marco verd... etiqueta de ma... estilo negrita
plotnLI4 = ReImPlot[Evaluate[e[ξ] /. soli4], {ξ, 0, NL[4]},
  represent... evalúa
  PlotLegends → Automatic, ReImLabels → {"Re E(ξ) interfase", "Im E(ξ) interfase"},
  leyendas de rep... automático etiquetas de parte... p... número e p... número e
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14}, Frame → True,
  estilo base familia de tipo de letra tamaño de tipo de l... marco verdadero
  FrameLabel → Style[ξ, 16, Bold], PlotStyle → {Blend[{Red, White}], Thick]];
  etiqueta de ma... estilo negrita estilo de repre... mezcla... rojo blanco grueso
Show[plotn1I4, plotnLI4, plotn2I4]
muestra

```

Out[*]=



```

In[ ]:= plotn1I4Abs = Plot[Abs[Evaluate[e[ξ] /. solI4]], {ξ, ξ1, 0},
    BaseStyle → {FontFamily → "CambriaMath", FontSize → 14}, Frame → True,
    FrameLabel → {Style[ξ, 16, Bold], Style["|E(ξ)|", 16, Bold]},
    PlotRange → {{ξ1, ξLG[4]}, {0.3, 1.6}}, PlotLegends → {"|E(ξ)|"}];

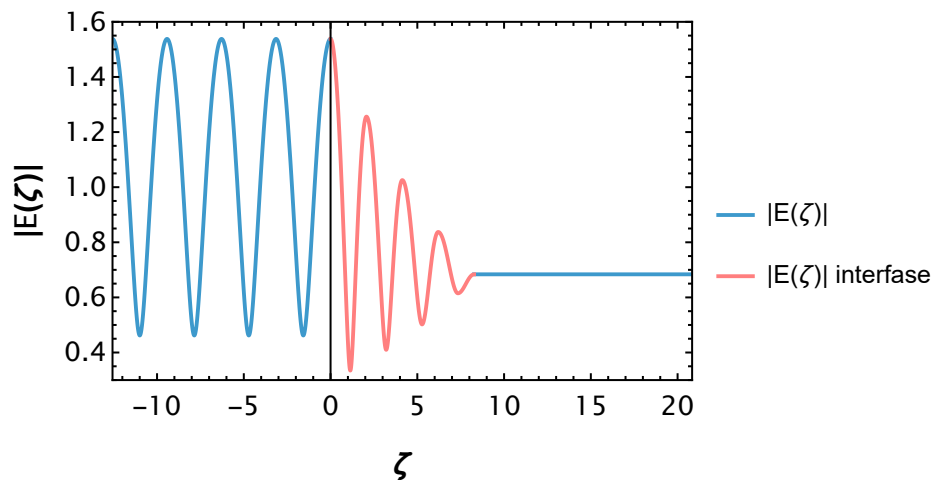
plotn2I4Abs = Plot[Abs[Evaluate[e[ξ] /. solI4]],
    {ξ, ξNL[4], ξLG[4]}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
    Frame → True, FrameLabel → {Style[ξ, 16, Bold], Automatic}];

plotnLI4Abs = Plot[Abs[Evaluate[e[ξ] /. solI4]],
    {ξ, 0, ξNL[4]}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
    Frame → True, FrameLabel → {Style[ξ, 16, Bold], Automatic},
    PlotStyle → {Blend[{Red, White}], Thick}, PlotLegends → {"|E(ξ)| interfase"}];

Show[plotn1I4Abs, plotn2I4Abs, plotnLI4Abs]

```

Out[]:=



```

In[ ]:= rI4num = (Evaluate[e[0] /. solI4] / E0) - 1

```

Out[]:=

```
{0.537921 - 0.000122244 i}
```

In[]:=

```
RI4num = Abs[rI4num] ^ 2
```

Out[]:=

```
{0.289359}
```

In[]:= **(*N=8*)**

In[*]:=

```
solI8 = NDSolve[{e''[ξ] + (nGgen[ξ, 8])^2 * e[ξ] == 0,
  I * n1 * e[ξ1] + e'[ξ1] - 2 * I * n1 * E0 * Exp[I * nGgen[ξ1, 8] * ξ1] == 0,
  I * nGgen[ξLG[8], 8] * e[ξLG[8]] - e'[ξLG[8]] == 0}, e, {ξ, ξ1, ξLG[8]}]
```

NDSolve: The scaled boundary value residual error of 4.694643250998806` indicates that the boundary values are not satisfied to specified tolerances. Returning the best solution found.

Out[*]=

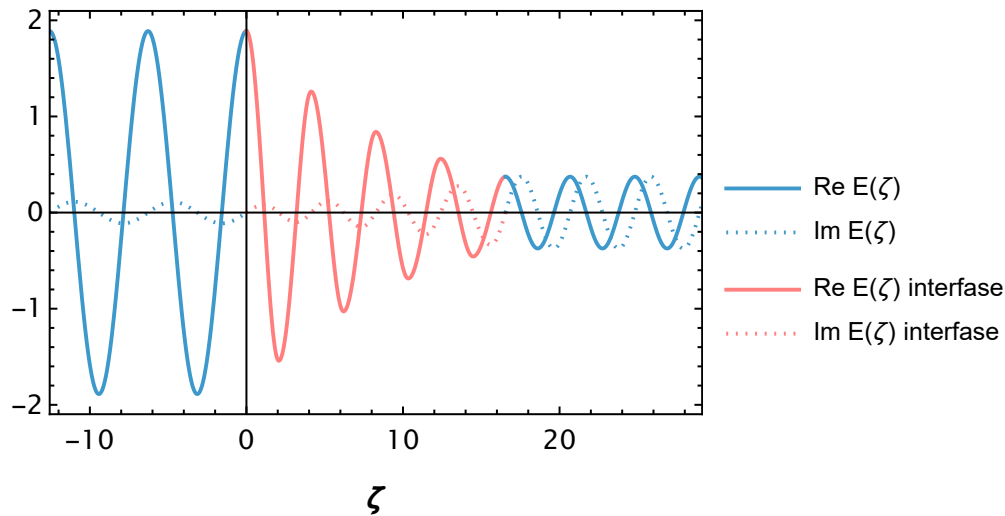
```
{ {e → InterpolatingFunction[ Domain: {{-12.6, 29.1}}
  Output: scalar ] ] }
```

```

In[*]:= plotn1I8 = ReImPlot[Evaluate[e[ $\zeta$ ] /. soli8], { $\zeta$ ,  $\zeta$ 1, 0}, PlotLegends → Automatic,
      represent... evalúa      leyendas de rep... automático
      ReImLabels → {"Re E( $\zeta$ )", "Im E( $\zeta$ )"}, PlotRange → {{ $\zeta$ 1,  $\zeta$ LG[8]}, Automatic},
      etiquetas de parte... p... número e p... número e rango de representación automático
      BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
      estilo base familia de tipo de letra tamaño de tipo de letra
      Frame → True, FrameLabel → Style[ $\zeta$ , 16, Bold]];
      marco verd... etiqueta de ma... estilo negrita
plotn2I8 = ReImPlot[Evaluate[e[ $\zeta$ ] /. soli8], { $\zeta$ ,  $\zeta$ NL[8],  $\zeta$ LG[8]},
      represent... evalúa
      BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
      estilo base familia de tipo de letra tamaño de tipo de letra
      Frame → True, FrameLabel → Style[ $\zeta$ , 16, Bold]];
      marco verd... etiqueta de ma... estilo negrita
plotnLI8 = ReImPlot[Evaluate[e[ $\zeta$ ] /. soli8], { $\zeta$ , 0,  $\zeta$ NL[8]},
      represent... evalúa
      PlotLegends → Automatic, ReImLabels → {"Re E( $\zeta$ ) interfase", "Im E( $\zeta$ ) interfase"},
      leyendas de rep... automático etiquetas de parte... p... número e p... número e
      BaseStyle → {FontFamily → "CambriaMath", FontSize → 14}, Frame → True,
      estilo base familia de tipo de letra tamaño de tipo de l... marco verdadero
      FrameLabel → Style[ $\zeta$ , 16, Bold], PlotStyle → {Blend[{Red, White}], Thick]];
      etiqueta de ma... estilo negrita estilo de repre... mezcla... rojo blanco grueso
Show[plotn1I8, plotnLI8, plotn2I8]
      muestra

```

Out[*]=



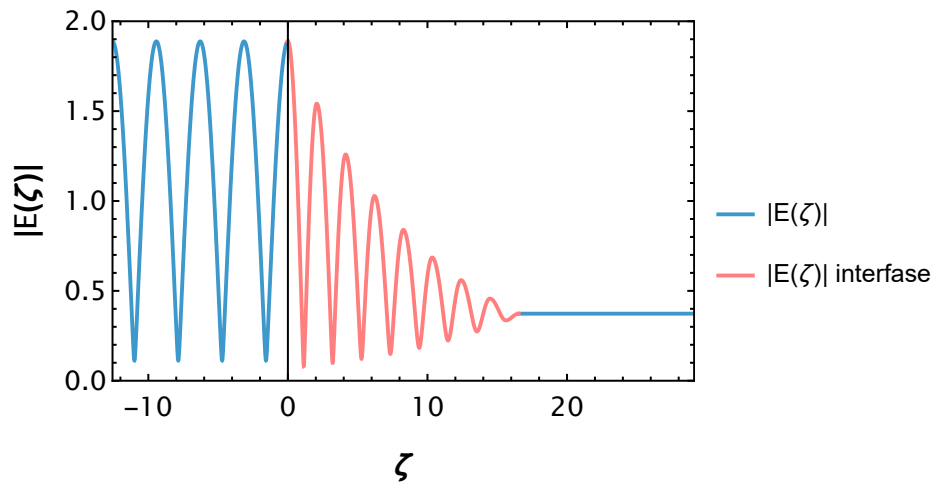
```

In[ ]:= plotn1I8Abs = Plot[Abs[Evaluate[e[ξ] /. solI8]], {ξ, ξ1, 0},
    [repr... [val... [evalúa
    BaseStyle → {FontFamily → "CambriaMath", FontSize → 14}, Frame → True,
    [estilo base [familia de tipo de letra [tamaño de tipo de letra [marco [verdadero
    FrameLabel → {Style[ξ, 16, Bold], Style["|E(ξ)|", 16, Bold]},
    [etiqueta de marco [estilo [negrita [estilo [número e [negrita
    PlotRange → {{ξ1, ξLG[8]}, {0, 2}}, PlotLegends → {"|E(ξ)|"}];
    [rango de representación [leyendas de represe... [número e
    plotn2I8Abs = Plot[Abs[Evaluate[e[ξ] /. solI8]],
    [repr... [val... [evalúa
    {ξ, ξNL[8], ξLG[8]}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
    [estilo base [familia de tipo de letra [tamaño de tipo de letra
    Frame → True, FrameLabel → {Style[ξ, 16, Bold], Automatic}}];
    [marco [verd... [etiqueta de marco [estilo [negrita [automático
    plotnLI8Abs = Plot[Abs[Evaluate[e[ξ] /. solI8]],
    [repr... [val... [evalúa
    {ξ, 0, ξNL[8]}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
    [estilo base [familia de tipo de letra [tamaño de tipo de letra
    Frame → True, FrameLabel → {Style[ξ, 16, Bold], Automatic},
    [marco [verd... [etiqueta de marco [estilo [negrita [automático
    PlotStyle → {Blend[{Red, White}], Thick}, PlotLegends → {"|E(ξ)| interfase"}];
    [estilo de repre... [mezcla... [rojo [blanco [grueso [leyendas de represe... [número e

    Show[plotn1I8Abs, plotn2I8Abs, plotnLI8Abs]
    [muestra

```

Out[]:=



```

In[ ]:= rI8num = ((Evaluate[e[0] /. solI8]) / E0) - 1
    [evalúa

```

Out[]:=

```
{0.887877 + 0.000078955 i}
```

```

In[ ]:= RI8num = Abs[rI8num]^2
    [valor absoluto

```

Out[]:=

```
{0.788326}
```

```
In[ ]:= (*N=16*)
```

```

In[*]:= solI16 = NDSolve[{e''[\xi] + (nGgen[\xi, 16])^2 * e[\xi] == 0,
    I * n1 * e[\xi1] + e'[\xi1] - 2 * I * n1 * E0 * Exp[I * nGgen[\xi1, 16] * \xi1] == 0,
    I * nGgen[\xiLG[16], 16] * e[\xiLG[16]] - e'[\xiLG[16]] == 0}, e, {\xi, \xi1, \xiLG[16]}]

```

 **NDSolve:** The scaled boundary value residual error of 20.854320137361377` indicates that the boundary values are not satisfied to specified tolerances. Returning the best solution found.

Out[*]=

```

{{e -> InterpolatingFunction[ Domain: {{-12.6, 45.7}}
    Output: scalar ]]}]}

```

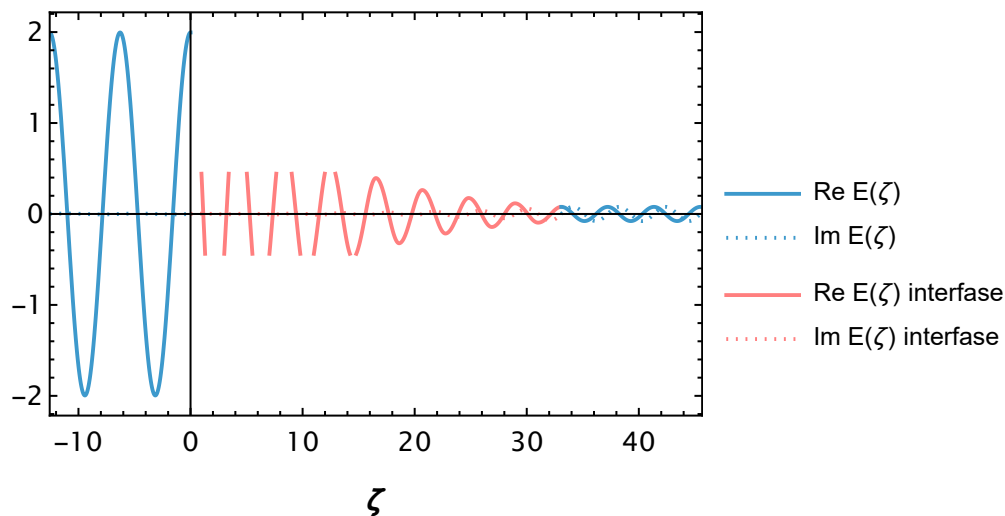


```

In[*]:= plotn1I16 = ReImPlot[Evaluate[e[ $\zeta$ ] /. solI16], { $\zeta$ ,  $\zeta$ 1, 0}, PlotLegends → Automatic,
      |represent... |evalúa |leyendas de rep... |automático
      ReImLabels → {"Re E( $\zeta$ )", "Im E( $\zeta$ )"}, PlotRange → {{ $\zeta$ 1,  $\zeta$ LG[16]}, Automatic},
      |etiquetas de parte... |p... |número e |p... |número e |rango de representación |automático
      BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
      |estilo base |familia de tipo de letra |tamaño de tipo de letra
      Frame → True, FrameLabel → Style[ $\zeta$ , 16, Bold]];
      |marco |verd... |etiqueta de ma... |estilo |negrita
plotn2I16 = ReImPlot[Evaluate[e[ $\zeta$ ] /. solI16], { $\zeta$ ,  $\zeta$ NL[16],  $\zeta$ LG[16]},
      |represent... |evalúa
      BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
      |estilo base |familia de tipo de letra |tamaño de tipo de letra
      Frame → True, FrameLabel → Style[ $\zeta$ , 16, Bold]];
      |marco |verd... |etiqueta de ma... |estilo |negrita
plotnLI16 = ReImPlot[Evaluate[e[ $\zeta$ ] /. solI16], { $\zeta$ , 0,  $\zeta$ NL[16]},
      |represent... |evalúa
      PlotLegends → Automatic, ReImLabels → {"Re E( $\zeta$ ) interfase", "Im E( $\zeta$ ) interfase"},
      |leyendas de rep... |automático |etiquetas de parte... |p... |número e |p... |número e
      BaseStyle → {FontFamily → "CambriaMath", FontSize → 14}, Frame → True,
      |estilo base |familia de tipo de letra |tamaño de tipo de l... |marco |verdadero
      FrameLabel → Style[ $\zeta$ , 16, Bold], PlotStyle → {Blend[{Red, White}], Thick]];
      |etiqueta de ma... |estilo |negrita |estilo de repre... |mezcla... |rojo |blanco |grueso
Show[plotn1I16, plotnLI16, plotn2I16]
|muestra

```

Out[*]=



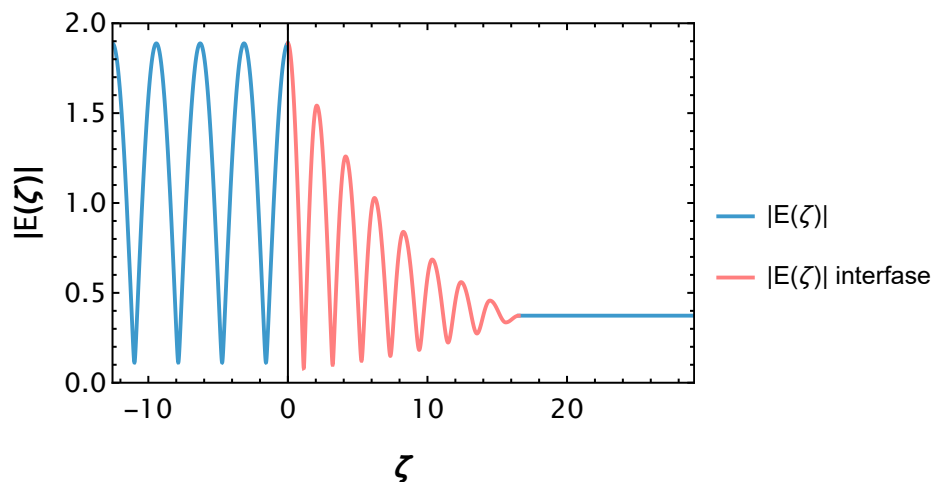
```

In[*]:= plotn1I16Abs = Plot[Abs[Evaluate[e[ξ] /. solI16]], {ξ, ξ1, 0},
    [repr...[val...[evalúa
    BaseStyle → {FontFamily → "CambriaMath", FontSize → 14}, Frame → True,
    [estilo base [familia de tipo de letra [tamaño de tipo de letra [marco [verdadero
    FrameLabel → {Style[ξ, 16, Bold], Style["|E(ξ)|", 16, Bold]},
    [etiqueta de marco [estilo [negrita [estilo [número e [negrita
    PlotRange → {{ξ1, ξLG[16]}, {0, 2}}, PlotLegends → {"|E(ξ)|"}];
    [rango de representación [leyendas de represe... [número e
plotn2I16Abs = Plot[Abs[Evaluate[e[ξ] /. solI16]],
    [repr...[val...[evalúa
    {ξ, ξNL[16], ξLG[16]}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
    [estilo base [familia de tipo de letra [tamaño de tipo de letra
    Frame → True, FrameLabel → {Style[ξ, 16, Bold], Automatic}}];
    [marco [verd... [etiqueta de marco [estilo [negrita [automático
plotnLI16Abs = Plot[Abs[Evaluate[e[ξ] /. solI16]],
    [repr...[val...[evalúa
    {ξ, 0, ξNL[16]}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
    [estilo base [familia de tipo de letra [tamaño de tipo de letra
    Frame → True, FrameLabel → {Style[ξ, 16, Bold], Automatic},
    [marco [verd... [etiqueta de marco [estilo [negrita [automático
    PlotStyle → {Blend[{Red, White}], Thick}, PlotLegends → {"|E(ξ)| interfase"}];
    [estilo de repre... [mezcla... [rojo [blanco [grueso [leyendas de represe... [número e

Show[plotn1I8Abs, plotn2I8Abs, plotnLI8Abs]
[muestra

```

Out[*]=



```

In[*]:= rI16num = ((Evaluate[e[0] /. solI16]) / E0) - 1
    [evalúa

```

Out[*]=

```
{0.99537 + 0.000174198 i}
```

```

In[*]:= RI16num = Abs[rI16num]^2
    [valor absoluto

```

Out[*]=

```
{0.990761}
```

In[*]:= **(*APARTADO J*)**

```
In[*]:=  $\rho[\text{Ncel\_}] := (n1 / n2) * (nL2 / nL1) ^ (2 * \text{Ncel}) ;$ 
```

```
In[*]:=  $R2 = ((1 - \rho[2]) / (1 + \rho[2])) ^ 2$ 
```

```
Out[*]=
```

```
0.0374289
```

```
In[*]:=  $R4 = ((1 - \rho[4]) / (1 + \rho[4])) ^ 2$ 
```

```
Out[*]=
```

```
0.289359
```

```
In[*]:=  $R8 = ((1 - \rho[8]) / (1 + \rho[8])) ^ 2$ 
```

```
Out[*]=
```

```
0.788325
```

```
In[*]:=  $R16 = ((1 - \rho[16]) / (1 + \rho[16])) ^ 2$ 
```

```
Out[*]=
```

```
0.990761
```